

Cucurbit Downy Mildew Update: July 30, 2025

Report of the cucurbit downy mildew status in Pennsylvania and surrounding states.

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Updated: July 29, 2025

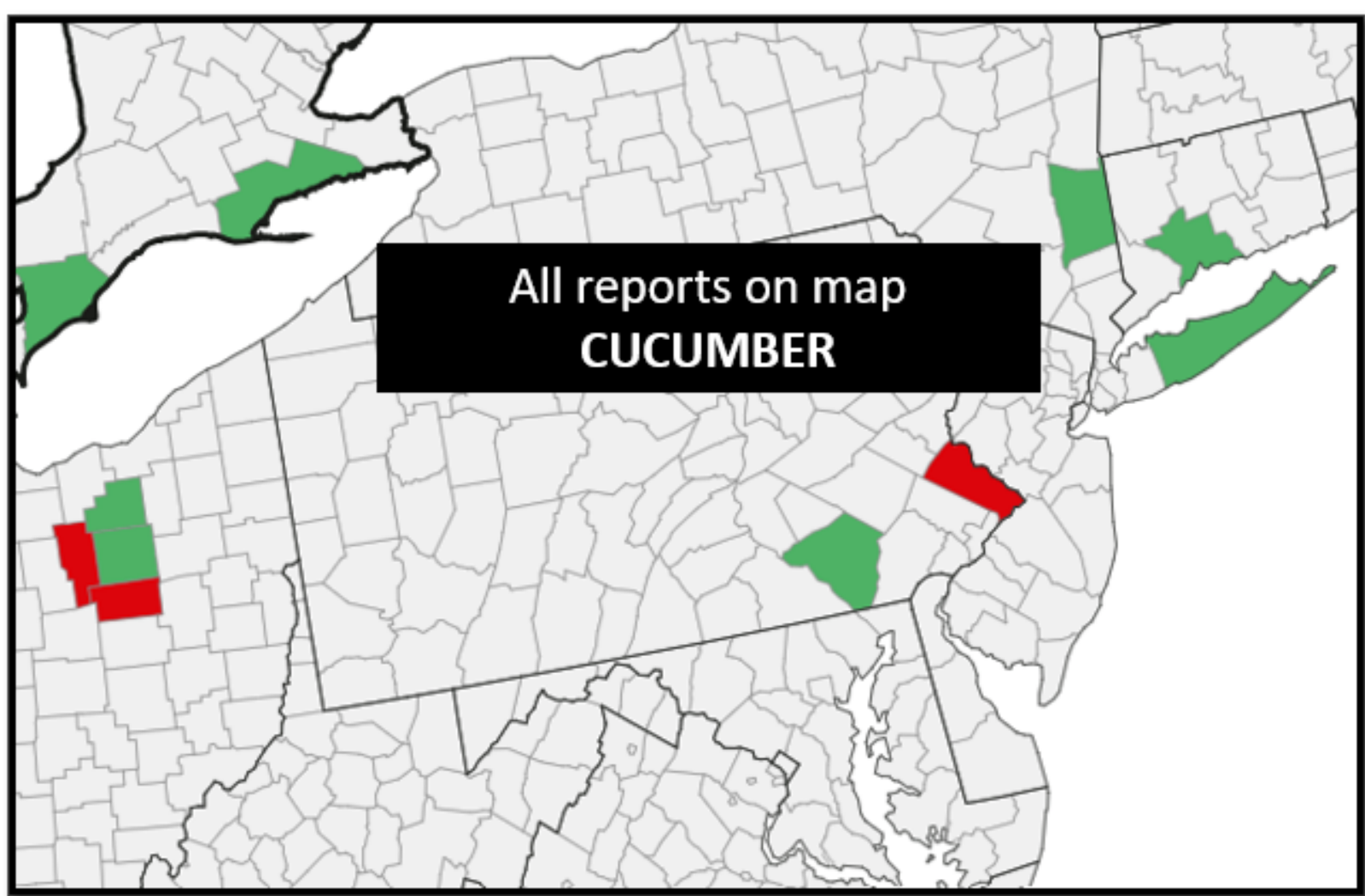


Image: Leaf symptoms of downy mildew. L. Fronk, Penn State

Downy mildew was reported on cucumber in Bucks County, PA on July 23, 2025.

Infections in Michigan, southern Pennsylvania, New York, Connecticut, and Ontario (map below) indicate that there are sources of the pathogen that can infect cucurbit crops coming from several directions, depending on the forecasted weather conditions. It is highly recommended that all cucumber and cantaloupe crops be protected with fungicides, even

those being grown in high tunnels. The high relative humidity in a high tunnel can be enough for downy mildew to develop. In order to be used in a high tunnel, the product must be labeled for greenhouse use. See Table E-12 in the Pest Management section of the [2024-2025 Mid-Atlantic Vegetable Production Recommendations](#) for a listing of products labeled for greenhouse production.



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Preventive applications are much more effective than applications made after the disease is detected. Since fungicides are the primary management tool for downy mildew, it is important to have a well-thought-out fungicide program that rotates among different FRAC codes for fungicide resistance management, as well as tank mixes with protectant fungicides that contain chlorothalonil (FRAC M05) and mancozeb (FRAC M03) if not already in the premix. Downy mildew targeted fungicides include, but are not limited to, Orondis Opti (FRAC 49 + M05), Ranman (FRAC 21), Elumin (FRAC 22), Zampro (FRAC 40 + 45), etc. See the [2024-2025 Mid-Atlantic Commercial Vegetable Production Recommendations](#) for a more complete list of fungicide recommendations.

Remember that fungicides that specifically target downy mildew will not manage powdery mildew.

Other foliar diseases that could be confused with downy mildew when scouting include angular leaf spot, anthracnose, Alternaria leaf blight, and powdery mildew.



Powdery mildew

Downy mildew is caused by the oomycete pathogen, *Pseudomonas cubensis*. Over the past several years, differential disease susceptibility between cucurbit hosts, as well as differential efficacy of targeted fungicides depending on the cucurbit host, have been observed and have led to dividing the pathogen into two clades. Pathogen isolates in one of the clades primarily affect cucumber and cantaloupe, while isolates from the other clade infect pumpkin, squash, and watermelon.

Extension educators are visiting farms and scouting for disease. Please email Leah Fronk at lf3339@psu.edu to contribute to this report. The next report will be posted on July 30, 2025.

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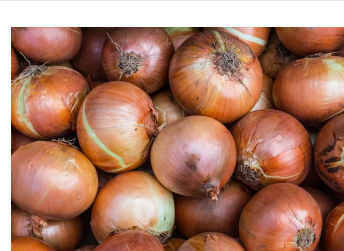
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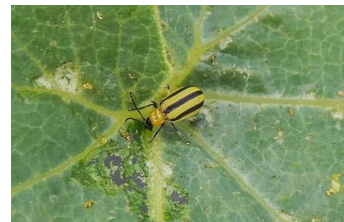
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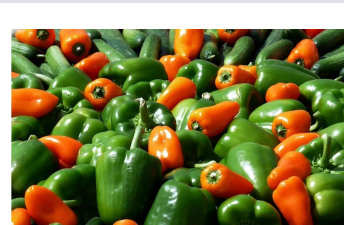
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